VISION-BASED SYSTEMS

Group Assessment: Vision-Based Landslide Forecasting Using Satellite Imagery

Module:	Vision-Based Systems
Assessment type:	Group project with individual evaluation
Group size:	3–5 students
Assessment commencement date:	[Insert date]
Final evaluation date:	[Insert date]
Repository submission deadline:	[Insert date]

[Department / Faculty / University]

[Academic Year / Semester]

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1. Background and Motivation

Landslides are among the most destructive natural hazards affecting Sri Lanka. They can cause loss of life, displacement of communities, destruction of houses, disruption of roads and utilities, and significant economic damage.

During the Ditwah Cyclonic Storm in 2025, prolonged and intense rainfall resulted in widespread landslides and slope failures across several areas of Sri Lanka. A reliable landslide forecasting system could support the early identification of vulnerable locations and contribute to disaster preparedness, evacuation planning, resource allocation, and the protection of human lives.

Accurately forecasting landslides is a difficult research problem because slope failures may be influenced by multiple factors, including terrain characteristics, vegetation, land use, soil conditions, geological characteristics, drainage, rainfall, moisture, and human activities. Nevertheless, modern satellite imagery, image-processing methods, machine learning, and deep-learning techniques provide opportunities to investigate visual patterns associated with locations that later experience landslides.

In this assessment, students are expected to investigate how theories and techniques learned in the Vision-Based Systems module can be applied to develop an experimental landslide forecasting system.

2. Dataset

The lecturer will provide a geographical dataset containing mapped landslide locations and boundaries associated with the Ditwah Cyclonic Storm in Sri Lanka in 2025.

The dataset is supplied in KML or KMZ format and contains geographical polygons representing mapped landslide or slope-failure areas. Students must extract and interpret the geographical information required for their proposed system.

The provided dataset identifies more than 4,000 mass movements and slope failures mapped using Sentinel-2 imagery and Remote Sensing and GIS techniques.

2.1 Dataset Acknowledgement

The landslide boundary demarcation dataset was prepared by the **Arthur C. Clarke Institute for Modern Technologies**, the nationally mandated institution for space science activities in Sri Lanka.

The dataset was prepared by:

- Mahesh Chathurange, Research Scientist, Space Applications Division;

- W.G.N.N. Jayawardhana, Research Scientist, Space Applications Division.

The dataset was prepared on behalf of the Arthur C. Clarke Institute for Modern Technologies.

Students must appropriately acknowledge the Arthur C. Clarke Institute for Modern Technologies in:

- the Git repository;
- project documentation;
- the final presentation;
- the prototype or user interface;
- any report, publication, demonstration, or other output developed using the dataset.

The dataset must be used only for academic, research, planning, and decision-support purposes. Students must not present the provided boundaries or the resulting model as legally or operationally verified information.

3. Main Assessment Objective

The objective of this assessment is to design, implement, and evaluate an experimental:

Vision-Based Landslide Forecasting System Using Satellite Imagery

Students must use the provided geographical landslide dataset to identify relevant locations and collect suitable satellite images corresponding to those locations.

The system should investigate whether visual characteristics observable in satellite imagery can be used to distinguish areas associated with later landslides from appropriate comparison areas.

Students are responsible for deciding:

- which satellite images should be collected;
- whether pre-event, post-event, multi-temporal, or composite images should be used;
- the appropriate date or date range;
- the geographical coverage of each image;
- whether the complete polygon, polygon centre, bounding region, or another sampling strategy should be used;

- how non-landslide or comparison samples should be selected;
- the image resolution and dimensions;
- the preprocessing methods;
- the model architecture;
- the training strategy;
- the evaluation metrics;
- the system design;
- the limitations and intended interpretation of the resulting model.

The lecturer may provide technical guidance and examples of suitable data-access services. However, selecting and justifying the final technical strategy is the responsibility of each student group.

4. Suggested Satellite-Image Services

Students may investigate suitable satellite imagery through services such as the following.

4.1 Copernicus Data Space Ecosystem

The Copernicus Data Space Ecosystem provides catalogue access, product downloads, visualisation, and processing services for Copernicus Sentinel data. Its available access methods include STAC, openEO, S3, and Sentinel Hub-related interfaces.

4.2 Microsoft Planetary Computer

Microsoft Planetary Computer provides a STAC API through which datasets such as Sentinel-2 Level-2A can be searched using geographical areas, dates, and other metadata.

4.3 Google Earth Engine

Google Earth Engine provides access to large geospatial datasets and cloud-based processing facilities. Noncommercial access requires project registration and verification and is subject to the applicable access conditions and quotas.

4.4 Other Services

Students may use other legally accessible and academically appropriate satellite-imagery services. However, they must:

- confirm that the imagery can be used for the intended academic purpose;
- record the image source;
- retain relevant metadata;
- comply with licensing and usage conditions;
- justify the suitability of the selected imagery.

Using Google Maps or Google Earth screenshots without checking the relevant licensing and data-usage conditions is not recommended.

5. Required Project Activities

The following stages provide the minimum expected workflow. Students may modify or extend the workflow where appropriate, provided all decisions are technically justified.

5.1 Stage 1: Discuss and Define the Project Strategy

Each group must first discuss and document its proposed forecasting strategy.

Students must not begin by selecting a model without first defining the problem and dataset strategy.

5.2 Stage 2: Acquire the Necessary Satellite Imagery

Students must develop or adopt a reproducible procedure to obtain satellite images corresponding to the geographical data supplied by the lecturer.

Students must justify why the selected images are suitable for forecasting.

The lecturer will not prescribe one compulsory image-acquisition strategy. Each group must make and defend its own decisions.

5.3 Stage 3: Prepare and Preprocess the Dataset

Students must apply suitable image-processing and dataset-preparation methods learned in the module.

Students must not apply techniques merely because they are available. Every major preprocessing decision must be justified and, where possible, experimentally evaluated.

The final dataset structure must be clearly documented.

At minimum, the dataset metadata should include:

- sample identifier;

- geographical coordinates;
- class label;
- associated landslide identifier, where applicable;
- satellite source;
- acquisition date or date range;
- image dimensions;
- spatial resolution;
- cloud information, where available;
- preprocessing operations applied.

5.4 Stage 4: Develop the Forecasting Model

Students must design and implement a vision-based model using concepts studied in the module.

Possible approaches may include:

- handcrafted visual features with a conventional classifier;
- Support Vector Machines;
- Random Forests;
- *k*-Nearest Neighbours;
- logistic regression;
- custom Convolutional Neural Networks;
- transfer learning;
- ensemble methods;
- temporal or multi-image approaches;
- other suitable computer-vision or machine-learning architectures.

The selection of the final model is entirely the responsibility of the students.

Students must be able to explain and defend:

- why the model was selected;
- the input representation;
- the architecture;
- the feature-extraction process;

- activation functions;
- loss function;
- optimizer;
- learning rate;
- batch size;
- number of epochs;
- regularization;
- augmentation;
- class-balancing strategy;
- training, validation, and testing procedure;
- hyperparameter choices;
- stopping conditions.

At least one meaningful baseline should be developed so that the group can demonstrate whether the proposed improvements provide a measurable benefit.

5.5 Stage 5: Evaluate Model Performance

Students must evaluate the system using suitable quantitative and qualitative methods.

Depending on the selected task, evaluation may include:

- accuracy;
- balanced accuracy;
- precision;
- recall;
- specificity;
- F1-score;
- false-positive rate;
- false-negative rate;
- ROC-AUC;
- confusion matrix;
- visualisation of correct and incorrect predictions;
- error analysis;

- model-comparison results.

Accuracy alone is not sufficient.

Students must explain the practical consequences of:

- false-positive forecasts;
- false-negative forecasts;
- class imbalance;
- geographical bias;
- seasonal bias;
- cloud contamination;
- insufficient image resolution;
- possible errors in the supplied inventory;
- uncertainty in the definition of non-landslide locations.

The evaluation must use a clearly defined training, validation, and testing strategy.

Students should take appropriate steps to prevent data leakage. Images from the same location, nearby overlapping regions, or the same landslide cluster should not be distributed across training and testing sets in a way that produces misleading results.

5.6 Stage 6: Develop a Continuous-Use System — Optional

Groups may develop a proper software system through which users can continuously apply the developed model.

Possible optional features include:

- map-based location selection;
- automatic satellite-image retrieval;
- model inference;
- forecast or susceptibility display;
- confidence score;
- visual explanation;
- history of analysed locations;
- report generation;
- image and metadata export;

- user authentication;
- administrator interface;
- interactive map visualisation.

The user interface must be relevant to the developed model and should not be treated as a substitute for a technically valid model.

6. Group Formation and Individual Responsibility

Each group must contain **three to five students**.

Although the project is completed as a group, every member will be evaluated individually.

Individual evaluation will consider:

- weekly progress;
- Git contributions;
- assigned responsibilities;
- technical contribution;
- familiarity with the complete system;
- conceptual understanding;
- ability to explain the code;
- ability to justify design decisions;
- ability to interpret results;
- contribution during the presentation and demonstration;
- responses during individual questioning.

Students within the same group may receive different marks.

Each student must understand the complete workflow, even where individual members have taken responsibility for particular components.

7. Git Repository and Continuous Progress

Each group must create a Git repository from the first day on which the assessment is introduced.

The repository must be shared with the lecturer in charge.

Students must push meaningful and traceable updates throughout the project. Repository activity should demonstrate continuous development rather than a large upload immediately before the final evaluation.

The repository should include:

- source code;
- configuration files;
- preprocessing scripts;
- data-acquisition scripts;
- model definitions;
- training scripts;
- evaluation scripts;
- interface code, where applicable;
- README file;
- installation and execution instructions.

Large satellite-image datasets, model weights, API keys, passwords, and private credentials must not be committed directly to a public Git repository.

Commit messages must meaningfully describe the changes made.

Examples of Unsuitable Commit Messages

```
update
changes
final
new file
done
```

Examples of Meaningful Commit Messages

```
Add KML polygon extraction and centroid calculation
Implement Sentinel-2 image search by date and cloud cover
Add geographically separated dataset split
Compare baseline CNN with transfer-learning model
Correct class weighting in training pipeline
Add confusion matrix and false-negative analysis
```

Each member should make identifiable contributions through their own account.

8. Weekly Progress Evaluation

At least **20 minutes of lecture time** will be allocated for evaluating project progress.

Progress evaluation may include:

- repository review;
- examination of recent commits;
- demonstration of completed work;
- explanation of technical decisions;
- inspection of the current dataset;
- discussion of difficulties;
- individual questioning;
- review of planned next steps.

The outcome of the weekly progress evaluation will contribute marks to the project.

Students must therefore begin work from the first day of the assessment and maintain regular progress.

A group that produces a final system without demonstrating continuous, independently performed development may receive reduced marks.

9. Final Assessment

The final assessment will include:

1. A working prototype of the developed model or forecasting system.
2. A short technical presentation.
3. A presentation of performance results.
4. Justification of major design decisions.
5. Individual questioning of group members.

The presentation should explain:

- the forecasting problem;
- the dataset;
- image-selection strategy;

- positive and negative sample selection;
- preprocessing;
- selected model;
- architecture;
- training procedure;
- evaluation;
- comparison with a baseline;
- important errors;
- limitations;
- possible improvements;
- potential practical use.

Students must demonstrate that the submitted results can be reproduced.

The detailed marking rubric for the final evaluation will be published on the date specified by the lecturer.

10. Competitive Model-Performance Marks

A total of **15 marks (out of 100)** will be allocated based on the comparative predictive performance of the developed models.

Subject to the validity of the submitted models and evaluation procedure:

- the group with the highest valid model performance will receive **15 marks**;
- the group with the second-highest valid performance will receive **10 marks**;
- the group with the third-highest valid performance will receive **5 marks**.

The ranking will be based on the common evaluation procedure and performance metric specified by the lecturer.

A model may be excluded from the competitive ranking where there is evidence of:

- training and testing data leakage;
- use of test data during model development;
- invalid or inconsistent class definitions;
- manipulation of the evaluation set;
- fabricated results;

- irreproducible results;
- use of unapproved external labels;
- other technically invalid practices.

The evaluation process will be transparent.

Students may challenge an awarded mark by submitting a clear technical justification supported by reproducible evidence.

The final decision will remain with the lecturer in charge.

11. Use of Large Language Models and Generative AI

Students may use Large Language Models and generative AI tools for activities such as:

- brainstorming;
- exploring possible approaches;
- receiving explanations;
- identifying possible errors;
- generating initial code suggestions;
- improving documentation;
- debugging assistance;
- researching alternative models.

However, AI tools must support student learning rather than replace it.

Students are responsible for:

- deciding the system architecture;
- selecting the image-acquisition strategy;
- selecting the preprocessing methods;
- selecting and modifying the model;
- verifying all generated code;
- evaluating model validity;
- checking references and factual claims;
- understanding every major component;
- ensuring that the work is reproducible;

- acknowledging significant use of AI tools.

Every student must have a conceptual understanding of the theories, techniques, algorithms, and models used in the project.

Every student must also be sufficiently familiar with the codebase to explain, modify, debug, and defend it.

Heavy or inappropriate dependence on LLM-generated work, external outsourcing, or copied implementations may result in a significant mark reduction.

Depending on the severity, the reduction may range from **10% to 90%** of the awarded project marks.

Examples of unacceptable conduct include:

- submitting code that group members cannot explain;
- being unable to modify or debug submitted code;
- generating a complete system without understanding its architecture;
- presenting fabricated model results;
- copying another group's work;
- outsourcing substantial parts of the project;
- concealing external assistance;
- claiming AI-generated decisions as independently justified decisions;
- failing to reproduce submitted results.

Serious cases may also be handled according to the applicable university academic-integrity procedures.

12. Minimum Deliverables

Each group must submit or demonstrate:

1. Shared Git repository.
2. Project strategy and problem definition.
3. Satellite-image acquisition method.
4. Dataset-generation and preprocessing code.
5. Dataset metadata.
6. Positive and comparison-sample strategy.

7. Baseline model.
8. Proposed forecasting model.
9. Training and validation procedure.
10. Final test results.
11. Confusion matrix and relevant performance metrics.
12. Error and limitation analysis.
13. Working prototype.
14. Short final presentation.
15. README and reproduction instructions.
16. Dataset and service acknowledgements.
17. Declaration of AI tools and external resources used.
18. Optional continuous-use system or user interface, where developed.

13. Important Scientific and Ethical Considerations

The resulting model must be presented as an experimental academic system.

Students must not claim that the model is a certified or operational disaster-warning system.

Satellite imagery alone may not provide all information required for reliable landslide forecasting. A real operational forecasting system may also require:

- rainfall measurements and forecasts;
- soil-moisture information;
- elevation and slope;
- geology;
- soil type;
- drainage;
- historical landslide information;
- land use;
- road-cut and construction information;
- field observations;

- validation by geologists and disaster-management professionals.

Students must clearly state the scope and limitations of their proposed system.

Model outputs must not be used to create unnecessary public alarm or make unsupported claims about the safety of a particular location.